

Practical Week 9

Topology

Task 89

```
SW1#config t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#vtp domain cisco
Domain name already set to cisco.
SW1(config)#vtp mode server
Device mode already VTP SERVER.
SW1(config)# vtp password hm322
Setting device VLAN database password to hm322

Enter configuration commands, one per line. End with CNTL/Z.
SW2(config)# vtp domain cisco
Changing VTP domain name from NULL to cisco
SW2(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to cisco

SW2(config)#vtp mode server
Device mode already VTP SERVER.
SW2(config)# vtp password hm322
Setting device VLAN database password to hm322

Enter configuration commands, one per line. End with CNTL/Z.
SW3(config)# vtp domain cisco
Changing VTP domain name from NULL to cisco
SW3(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to cisco.

SW3(config)# vtp mode server
Device mode already VTP SERVER.
SW3(config)# vtp password hm322
Setting device VLAN database password to hm322
```

Task 90

VLAN Number	10
VLAN Name	Engineering
VLAN Number	20
VLAN Name	Sales

FastEthernet0/1	
Port Status	☒ On
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
Access	VLAN 10

FastEthernet0/2	
Port Status	☒ On
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
Access	VLAN 10

FastEthernet0/3		
Port Status	☒ On	
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto	
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto	
Access	VLAN	20

FastEthernet0/1		
Port Status	☒ On	
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto	
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto	
Access	VLAN	20

FastEthernet0/2		
Port Status	☒ On	
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto	
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto	
Access	VLAN	20

FastEthernet0/3		
Port Status	☒ On	
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto	
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto	
Access	VLAN	10

Task 91

GigabitEthernet0/1		
Port Status	☒ On	
Link Speed	☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto	
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto	
Trunk	VLAN	2-1001

GigabitEthernet0/2		
Port Status	☒ On	
Link Speed	☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto	
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto	
Trunk	VLAN	2-1001

Task 92

SW1#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
10	Engineering	active	Fa0/1, Fa0/2
20	Sales	active	Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

SW1#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	1
Port	Vlans allowed on trunk			
Gig0/1	2-1001			
Port	Vlans allowed and active in management domain			
Gig0/1	10,20			
Port	Vlans in spanning tree forwarding state and not pruned			
Gig0/1	10,20			

SW1#

SW2#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	auto	n-802.1q	trunking	1
Gig0/2	on	802.1q	trunking	1
Port	Vlans allowed on trunk			
Gig0/1	2-1001			
Gig0/2	2-1001			
Port	Vlans allowed and active in management domain			
Gig0/1	10,20			
Gig0/2	10,20			
Port	Vlans in spanning tree forwarding state and not pruned			
Gig0/1	10,20			
Gig0/2	10,20			

SW3#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1
10	Engineering	active	Fa0/3
20	Sales	active	Fa0/1, Fa0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

SW3#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gig0/2	auto	n-802.1q	trunking	1
Port	Vlans allowed on trunk			
Gig0/2	2-1001			
Port	Vlans allowed and active in management domain			
Gig0/2	10,20			
Port	Vlans in spanning tree forwarding state and not pruned			
Gig0/2	10,20			

Task 93

FastEthernet0/1	
Port Status	☒ On
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
Access ▼	VLAN 10 ▼

FastEthernet0/2	
Port Status	☒ On
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
Access ▼	VLAN 20 ▼

FastEthernet0/0	
Port Status	☒ On
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	000C.85D1.1701
IP Configuration IPv4 Address 10.10.10.1 Subnet Mask 255.255.255.0	

FastEthernet0/1	
Port Status	☒ On
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	000C.85D1.1702
IP Configuration IPv4 Address 10.10.20.1 Subnet Mask 255.255.255.0	

```
SW2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24
10	Engineering	active	Fa0/1
20	Sales	active	Fa0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Task 94

```
R1(config)# interface fa0/0.10
R1(config-subif)# encapsulation dot1q 10
R1(config-subif)# ip address 10.10.11.1 255.255.255.0
R1(config-subif)# exit
R1(config)# interface fa0/0.20
R1(config-subif)# encapsulation dot1q 20
R1(config-subif)# ip address 10.10.21.1
% Incomplete command.
R1(config-subif)# ip address 10.10.21.1 255.255.255.0

R1#show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
FastEthernet0/0          10.10.10.1      YES manual up            up
FastEthernet0/0.10       10.10.11.1      YES manual up            up
FastEthernet0/0.20       10.10.21.1      YES manual up            up
FastEthernet0/1          10.10.20.1      YES manual up            up
Vlan1                    unassigned      YES unset  administratively down down

SW2#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Gig0/1    auto      n-802.1q       trunking      1
Gig0/2    on        802.1q         trunking      1

Port      Vlans allowed on trunk
Gig0/1    10,20
Gig0/2    2-1001

Port      Vlans allowed and active in management domain
Gig0/1    10,20
Gig0/2    10,20

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    10,20
Gig0/2    10,20
```

Task 95

```
SW2(config)# ip routing
SW2(config)# interface vlan 10
SW2(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

SW2(config-if)# ip address 10.10.10.1 255.255.255.0
SW2(config-if)# no shutdown
SW2(config-if)# exit
SW2(config)# ip routing
SW2(config)# interface vlan 20
SW2(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

SW2(config-if)# ip address 10.10.20.1 255.255.255.0
SW2(config-if)# no shutdown

Vlan1                    unassigned      YES unset  administratively down down
Vlan10                   10.10.10.1      YES manual up            up
Vlan20                   10.10.20.1      YES manual up            up
```

Task 96

Configuring separate interfaces on a router would work best because a campus has many computers to be configured.